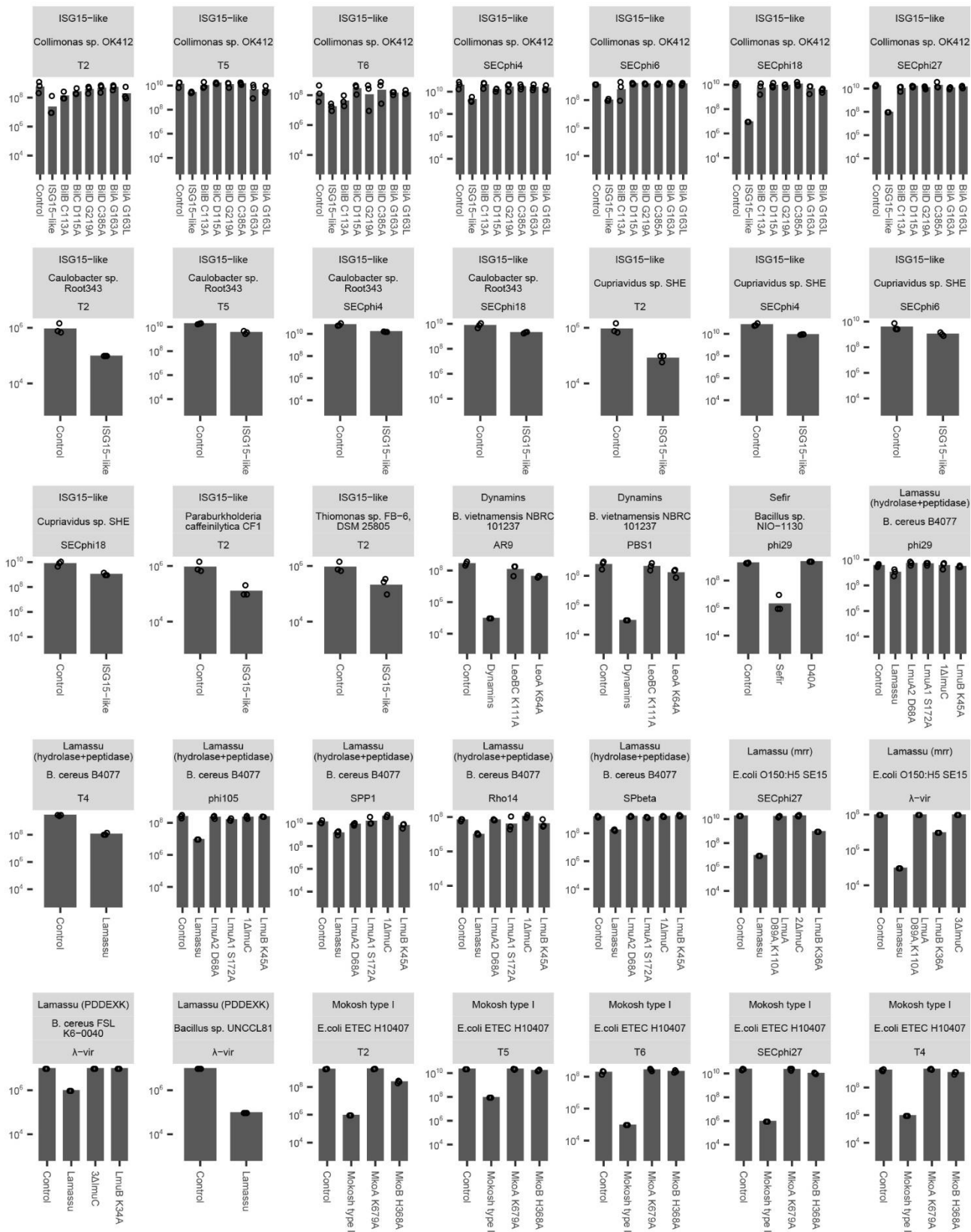
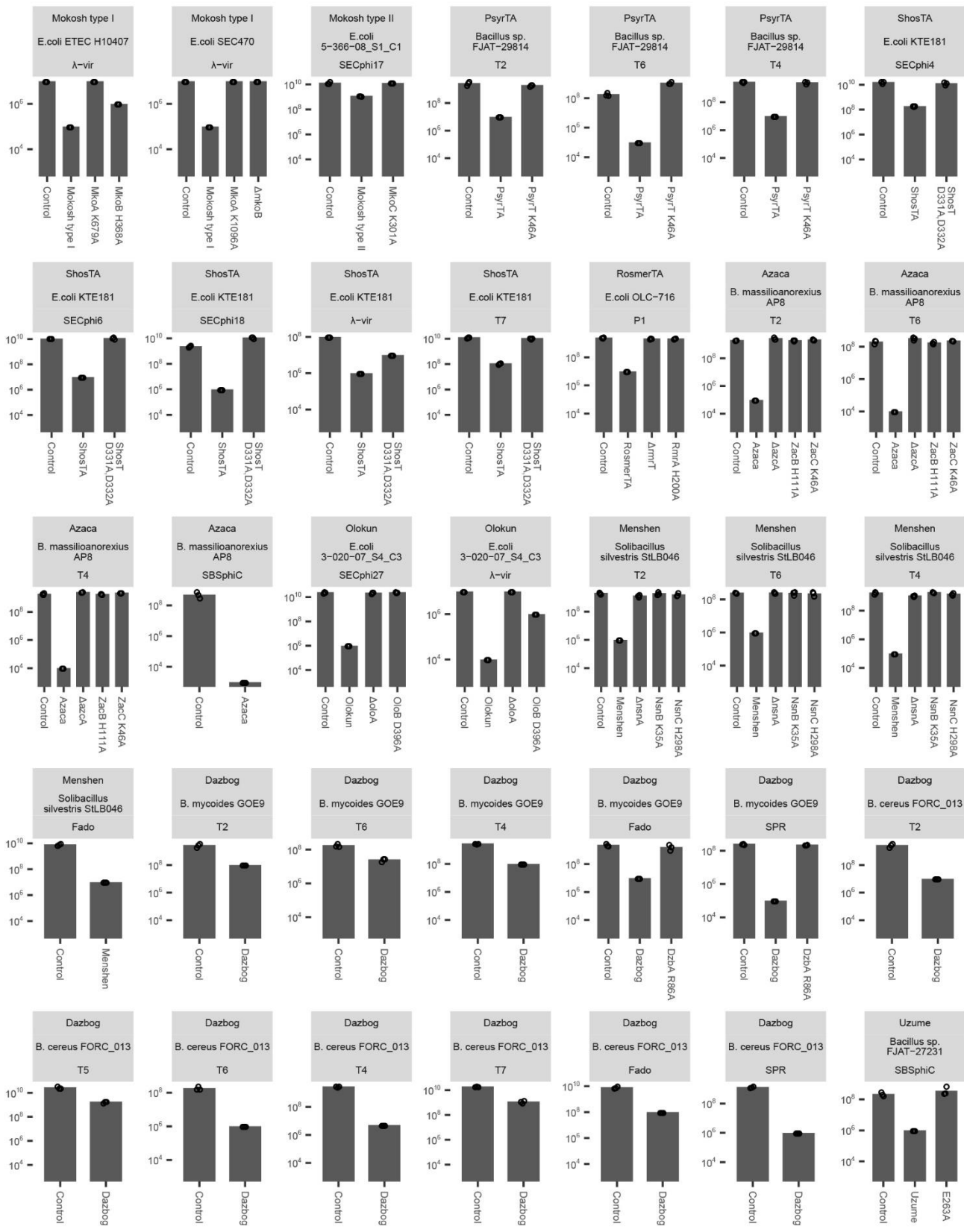


Supplemental information

**An expanded arsenal of immune systems
that protect bacteria from phages**

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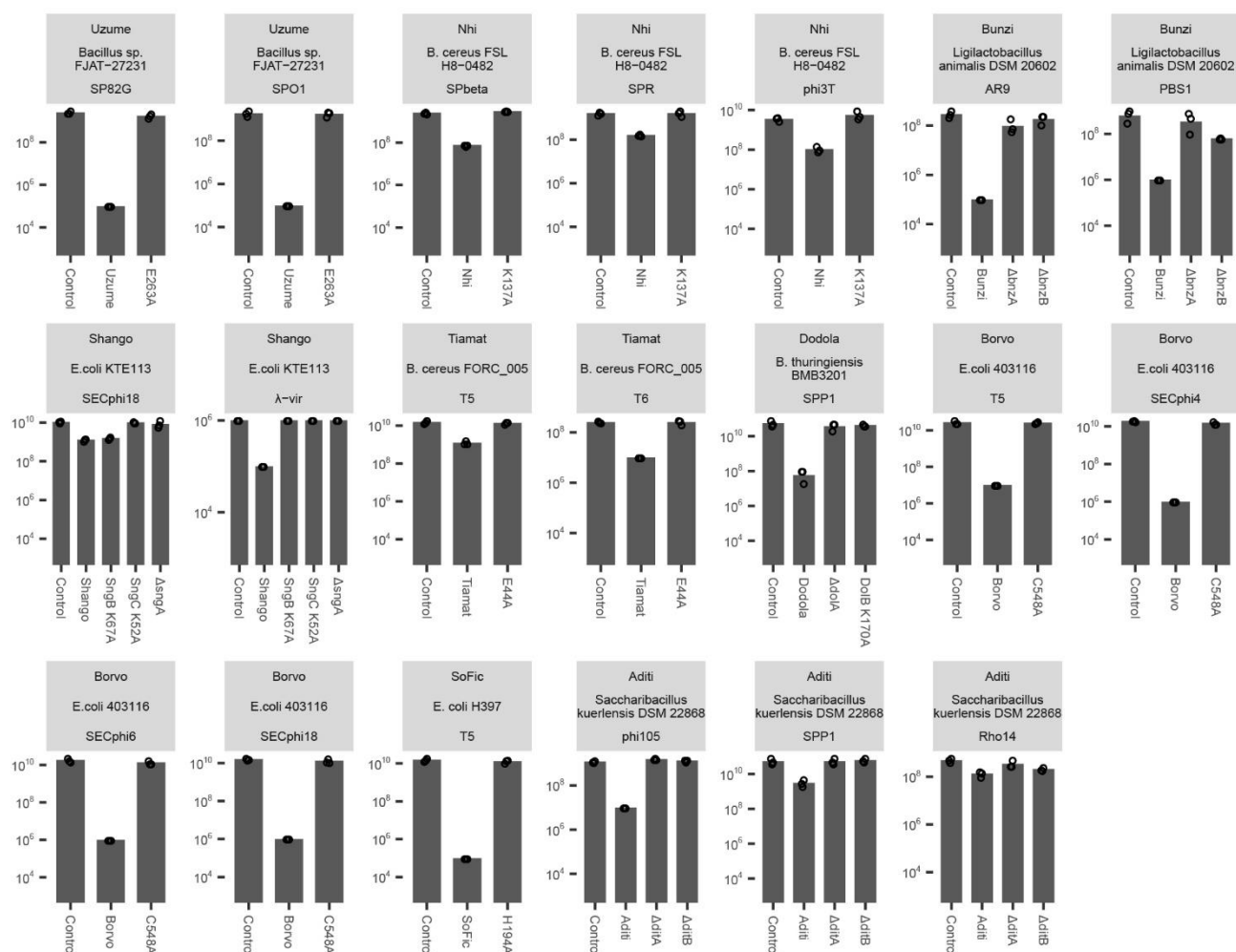


Figure S2. Plaque assay results for all defensive systems described in this study, related to Figures 1-4.

Efficiency of plating (EOP) of phages infecting *E. coli* or *B. subtilis* with and without WT and mutated systems. Data shown only for phages for which defense was detected. Data represent plaque-forming units (PFU) per ml; bar graphs represent average of three replicates, with individual data points overlaid.

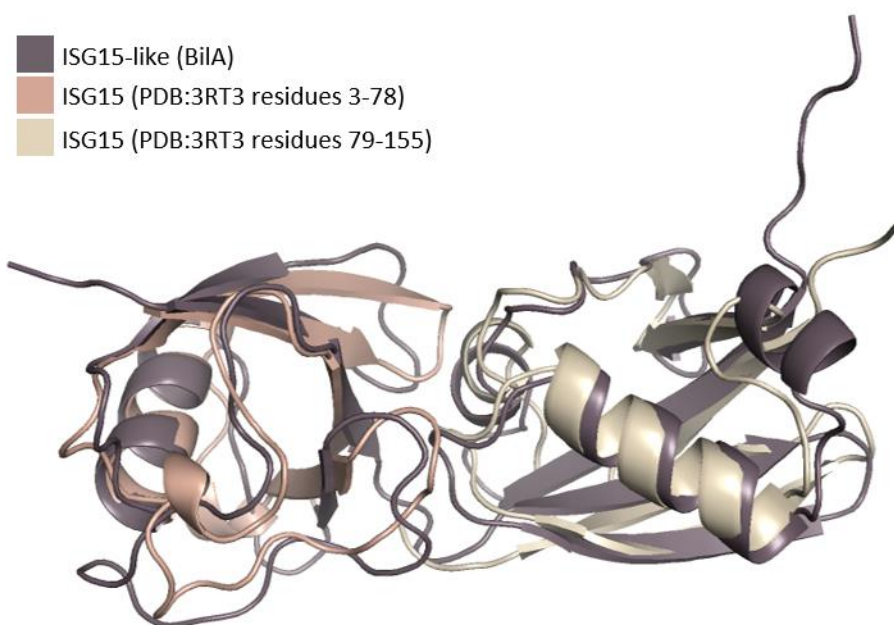


Figure S3. Structural alignment of the ISG15-like protein BilA from *Collimonas sp.* OK412 and the human ISG15 (PDB:3RT3 chain B) , related to Figure 1.

Each ubiquitin-like domain from ISG15 was aligned separately. Alignment was performed using PDBeFold (Krissinel and Henrick, 2004) and visualized using PyMOL (Schrödinger, 2015).

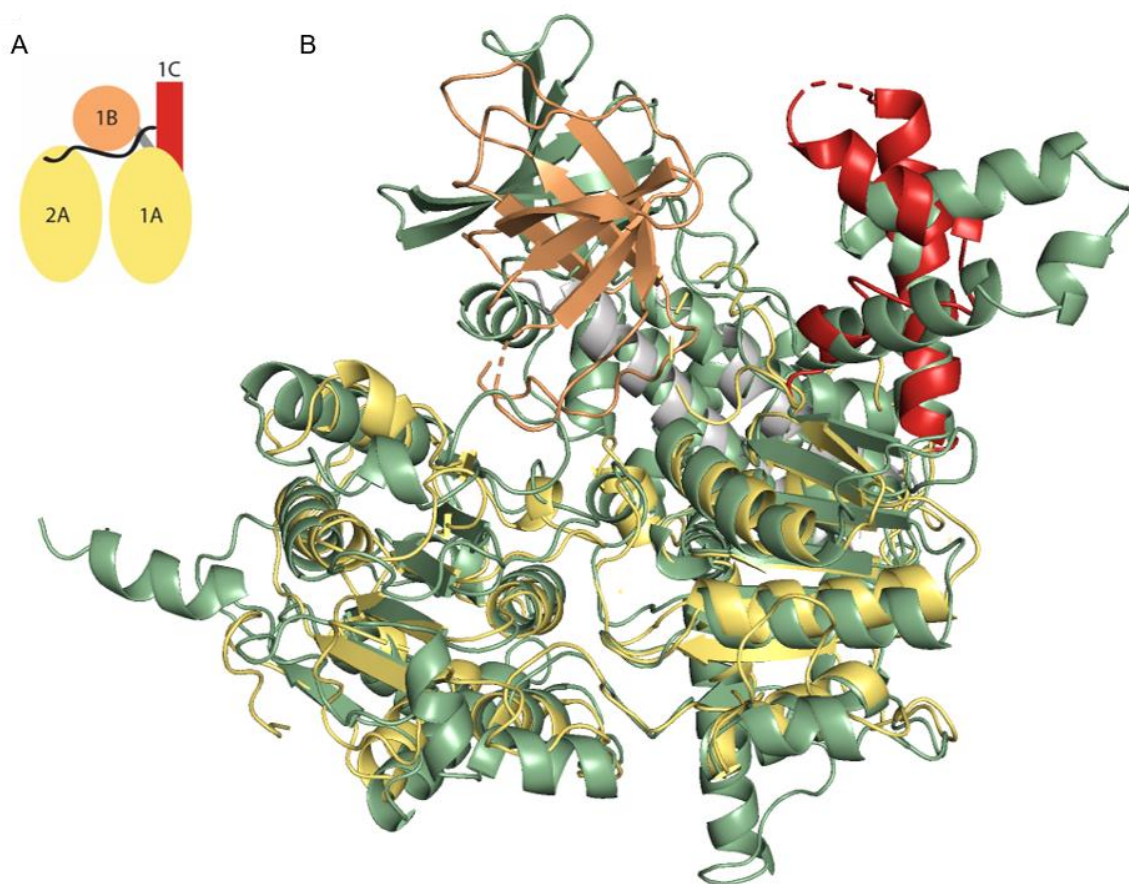


Figure S4. Structural homology of Mokosh protein MkoA to the human Upf1 helicase, related to Figure 3.

(A) Schematic representation of Upf1 family helicases. The Upf1 protein has two RecA domains (yellow) and two additional domains (orange and red). Domains are colored as in (Chakrabarti et al., 2011; Gowravaram et al., 2018). (B) Structural alignment of the AlphaFold-predicted structure of the helicase domain (residues 460-1215) of Mokosh MkoA protein from *E. coli* ETEC H10407 (green) and Upf1 (PDB:6EJ5).



Figure S5. Phylum-specific distribution of defense systems, related to Figure 5.

X-axis represents the number of defense systems per genome; Y-axis represents the fraction of genomes within the phylum. Dashed line is the average out of the entire set of 3,895 genomes; Solid gray line is the average within the phylum. Only phyla with at least 10 finished genomes in our database are presented. Some of the data presented here are also presented in Figure 5.

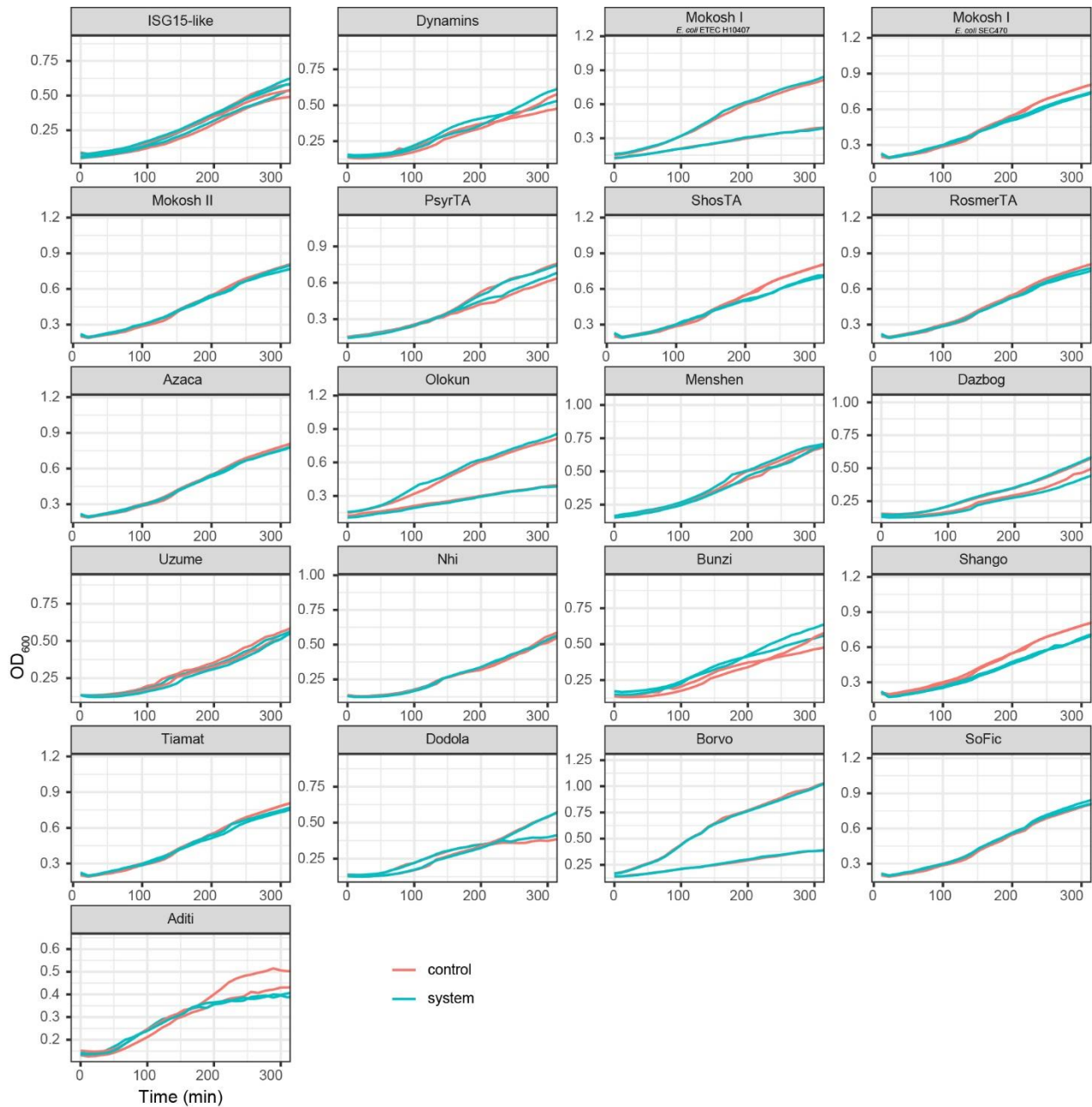


Figure S6. Growth curves of system-containing and system-lacking cells in liquid culture, related to Figures 1-4.

Growth curves for *E. coli* or *B. Subtilis* cells harboring a plasmid with the mentioned system (blue) or an empty vector (red). Curves represent two biological replicates for each system. All experiments were done at room temperature.